

279kWh **Smart** Battery Cabinet



279^{kWh}
NAMEPLATE
ENERGY

125^{kW}
GUARANTEED
MIN POWER

821^V
NOMINAL
VOLTAGE

6K⁺
CYCLE LIFE
@ 80% DOD

-40/+55^{°C}
OPERATING
TEMP

IP54
ENCLOSURE
RATING

20^{yr}
DESIGN
LIFE

Smart
ENERGIOS™
ENABLED

Series B 279kWh Smart Battery



- Sodium-Ion NFPP
- Air-Cooled Cabinet
- EnergiOS™ Smart EMS
- IP54 Outdoor-Rated



Built for critical infrastructure resiliency and demanding commercial & industrial use cases, the Series B 279kWh Smart Battery Cabinet prioritizes compliance, field-validated performance, and bankable warranties — delivering a **20-year design life** and **6,000+ cycles at 80% DoD** with zero maintenance and no HVAC required.

INCLUDES AN OUTDOOR, IP54-RATED ENCLOSURE CONTAINING:

- Sodium-Ion NFPP battery modules
- Air-cooled thermal management
- Auxiliary safety systems
- Battery Management System (BMS)
- EnergiOS™ Smart EMS platform
- Fire suppression & gas detection

KEY BENEFITS



Safety-First Design

Sodium-Ion NFPP chemistry eliminates thermal runaway risk — enabling safer siting, simpler AHJ approvals, and no fire suppression complexity.

- ✓ Zero thermal runaway incidents
- ✓ No toxic gas emissions
- ✓ Air-shippable — faster deployment
- ✓ Cleaner AHJ approvals



Extreme Performance

Operates from -40°C to $+55^{\circ}\text{C}$ without derating or HVAC infrastructure — ideal for outdoor installations, cold storage, and industrial environments.

- ✓ -40°C to $+55^{\circ}\text{C}$ operation
- ✓ No HVAC required
- ✓ Minimal capacity derating
- ✓ Fast charging capability



Intelligent & Bankable

Built-in EnergiOS™ Smart EMS transforms the cabinet into an active energy asset — optimizing dispatch, enabling VPP participation, and maximizing ROI.

- ✓ 6,000+ cycles @ 80% DoD
- ✓ 20-year design life
- ✓ ITC eligible with solar PV
- ✓ Demand response & VPP ready

WHERE THE SERIES B PERFORMS



MANUFACTURING / COLD STORAGE / DISTRIBUTION

Peak Demand Management

Reduce demand charges 30–50% at manufacturing facilities, cold storage, and distribution centers while maintaining full capacity in extreme temperatures.



DATA CENTERS / HEALTHCARE / TELECOM

Critical Infrastructure

Guaranteed uptime for data centers, hospitals, and telecom — with air-cooled simplicity, intelligent monitoring, and no fire suppression complexity.



WAREHOUSE / PORT / AIRPORT

Material Handling & Fleet

Fast charging and extreme temp tolerance for 24/7 operations — warehouse automation, port equipment, and airport ground support.



SOLAR / WIND / CAMPUS / COMMUNITY

Renewable Integration

Long-duration storage with 20-year design life for bankable solar + storage projects, campus microgrids, wind smoothing, and community solar.

THREE TECHNOLOGIES. ONE INTELLIGENT PLATFORM.

Not Just a Cabinet — An Intelligent Energy Platform

The Series B integrates with Coulomb Technology's Smart EMS platform — a suite of three proprietary software technologies that transform your battery cabinet into an active, revenue-generating energy asset. Monitor, optimize, and control your energy in real time from any device.



The intelligent operating system for your energy storage. Provides real-time monitoring, automated charge/discharge scheduling, and seamless solar integration.

- Real-time SOC, power flow & health monitoring
- Automated TOU & solar charge scheduling
- Remote access via web & mobile app
- Alerts, diagnostics & OTA firmware updates



Machine learning algorithms that learn energy usage patterns, weather forecasts, and utility rates to autonomously optimize dispatch and maximize savings.

- Predictive load forecasting & solar yield modeling
- Autonomous dispatch optimization
- Dynamic rate arbitrage (TOU, demand charges)
- Continuous learning — improves over time



Advanced grid interaction capabilities enabling virtual power plant participation, demand response programs, and utility incentive monetization.

- Virtual power plant (VPP) participation
- Demand response & utility incentive programs
- Grid frequency regulation support
- Export control & islanding management

INTELLIGENT BMS PROTECTIONS

Overcharge Protection

Overcurrent Protection

Overdischarge Protection

Overheat Protection

Temperature Management

Short Circuit Protection

Balanced Power

Reverse Connection

TOTAL COST OF OWNERSHIP ADVANTAGE

20_{yr}
DESIGN LIFE
VS. 3-5 YR LEAD-ACID

Zero
MAINTENANCE COST
OVER LIFETIME

6K+
CYCLES @ 80% DOD
VS. 300-500 LEAD-ACID

50%
DEMAND CHARGE
REDUCTION POTENTIAL



MODEL ASSUMPTIONS	
NFPP installed price delta	\$16/kWh
Cycles per year	365
HVAC load delta	90%
Insurance cost delta	20%
Augmentation delta	20%
O&M cost delta	30%
Cell derating / cold temps	75%
Discount rate / NPV calc	8%
COST COMPONENTS	
LFP — CapEx	NFPP — CapEx
LFP — HVAC Load	NFPP — O&M
LFP — O&M	NFPP — Insurance
LFP — Insurance	LFP — Augmentation

SERIES B — 279KWH
CABINET



NAMEPLATE ENERGY

279kWh

Nameplate energy capacity

GUARANTEED MIN POWER

125kW

Guaranteed minimum power output

CYCLE LIFE

6K+

At 80 % DoD, 25°C

NOMINAL VOLTAGE

821v

High-voltage DC bus

OPERATING TEMP

−40/+55°C

No HVAC — any climate

ENCLOSURE RATING

IP54

Outdoor-rated direct install

PARAMETER	VALUE / DESCRIPTION	UNIT
NOMINAL VOLTAGE	279 (high-voltage DC bus)	VDC
NAMEPLATE ENERGY CAPACITY	50	kWh
USABLE ENERGY (85% DOD)	42.5	kWh
MAX RATED POWER	25	kW
CYCLE LIFE	6,000+ at 80% DoD, 25°C	—
CELL TYPE	Sodium-Ion NFPP (Non-Flammable Polymer Pouch)	—
OPERATING TEMP (DISCHARGE)	−40°C to +55°C	—
CHARGING TEMP RANGE	−10°C to +45°C	—
DESIGN LIFE	20 years	—
ENCLOSURE (IP RATING)	Outdoor Cabinet — IP54	—
THERMAL MANAGEMENT	Air-Cooled (No HVAC Required)	—
BMS COMMUNICATION	Integrated BMS, Modbus TCP / CAN Bus	—
SMART EMS	EnergiOS™ — Real-time monitoring, TOU scheduling, OTA updates	—
AUXILIARY SYSTEMS	Fire Suppression, Gas Detection, Explosion Venting	—

CERTIFICATIONS

UL 1973

UL 9540

UN 38.3

IEC 62619

Cell & Module Level

UL 9540A TESTING

Cell: Tested & Passed

Module/System: In Progress

WARRANTY

Cell: 3 Years

System: 5 Years

Extended warranty available

FACTORY ACCEPTANCE TEST (FAT) SCOPE

- ✓ Visual, Mechanical & Torque Inspection

✓ 3 × 0.125C Capacity/Energy Verification (85% DOD)
- ✓ Protections & Isolation Tests

✓ Baseline OCV Measurement
- ✓ Functional & Conditioning Test

✓ 10 × 0.5C Stability Cycles with Power Checks